P3 REVOKE-HDC

Week 1: Freeze Update Alphabet, Residue and Record Format

Submission-Ready Reference Solution and Detailed Technical Manual

Revision: W1-R1
Date: 25 September 2026
Platform target: AMD Kria KV260 / Zynq UltraScale+ MPSoC
Contract status: complete; evidence status: AMBER pending team-specific empirical inputs

Purpose
This manual explains the entire Week 1 solution chronologically, including exact packet/journal formats, signed arithmetic, validation precedence, restoration ordering, memory sizing, refusal semantics, test vectors, hardware integration choices and the Friday acceptance review. It does not invent missing empirical evidence.

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Source truth boundary
The uploaded WEEK_1.md references README.md, PROJECT_PLAN.md and EVALUATION.md, but those files were not supplied. This manual marks source-frozen facts, R1 engineering decisions, and external evidence dependencies separately so the submission is technically usable without pretending that missing measurements were performed.

1. Scope, source boundary and final deliverables

Week 1 is a specification and evidence week. No simulator model or RTL is required. The goal is to define exactly what a sparse journal record can validate and later reverse, with enough numerical detail that another team can validate a sample packet and compute storage without verbal explanation.

1.1 What is frozen by the supplied brief

- 16-byte wire/header schema.
- 3-byte wire entry for each affected dimension.
- 2-byte persisted journal entry for each affected dimension.
- 4096 prototype dimensions (12-bit dimension index) and four prototypes/classes.
- Maximum F=96 affected dimensions per packet.
- 4-bit signed pre-update residue and 4-bit signed update contribution.
- Eleven outcome codes 0 through 10.
- Restoration reconstructs the old packed bit from retained residue; it does not reconstruct the hidden full signed count.

1.2 R1 engineering decisions needed to make the contract executable

- Little-endian multi-byte fields.
- Two's-complement signed nibbles.
- Zero tie convention: sign_threshold(0)=1.
- CRC-16/CCITT-FALSE parameters.
- No-split ring wrap with pad-to-end and restart at zero.
- 100 MHz PL clock target and 50k packet-decisions/s engineering target.
- 256 KiB journal ring, N=1024 index slots, 4096-entry trace.
- AXI-Lite project allocation at 0xA0000000 via M_AXI_HPM0_FPD -> SmartConnect.

1.3 Evidence that cannot be truthfully synthesized from the uploaded file

- Actual encoder and dataset revision/split.
- Measured per-dimension contribution distribution.
- Calibrated theta_A and theta_B.
- Repository Git commit/build ID.
- Independent human reviewer/countersignature.
- Exact controller state names from missing PROJECT_PLAN Section 3.3.

1.4 Deliverable package

Artifact	Purpose
README_SUBMISSION.md	entry point, final numbers, status
docs/interface_contract.md	wire/journal byte and bit contract
docs/validation_rules.md	equations, precedence, restoration math
docs/decision_log.md	all sixteen Week 1 decisions
docs/ordering_state.md	record lifecycle and controller flow
docs/sizing_sheet.md	record sizing and memory math
docs/refusal_table.md	outcome triggers 0..10
tests/hand_vectors.*	36 valid worked vectors + 6 refusals
configs/revoke_hdc_week1_v1.yaml	machine-readable frozen parameters
evaluation/EVALUATION_WEEK1.md	Friday review
manual/...pdf	this detailed manual

2. Executive answer: what one record can prove and reverse

A valid persisted record proves which prototype dimensions were affected and preserves enough pre-update information to reconstruct the packed bit that existed immediately before the accepted update. For each dimension it stores the 12-bit dimension index and the 4-bit signed pre-update residue. The old bit is recomputed as sign_threshold(residue).

```
old_bit = sign_threshold(pre_update_residue)
new_bit = sign_threshold(pre_update_residue + update_contribution)
sign_threshold(x) = 1 if x >= 0 else 0
```

The record intentionally does not store the hidden full count. Consequently, the reversible object is the 4096-bit packed prototype state, not Arm A's internal count vector. This distinction is central to correctness: the record can restore exactly the bit-state it is designed to restore without claiming to reconstruct information it never retained.

2.1 Core numeric results

Quantity	Result
Packed prototype	4096 bits = 512 B
Wire packet	L_wire(F)=16+3F B
Journal record	R(F)=16+2F B
Max F	96
Max supported journal record	R(96)=208 B
Break-even	16+2F<512 -> F<248
F=247	510 B, strictly smaller
F=248	512 B, equal; no strict storage win

Most important Friday number
The maximum supported persisted journal record is 208 bytes because the frozen packet cap is F=96. The F=247/F=248 calculations are theoretical break-even demonstrations only.

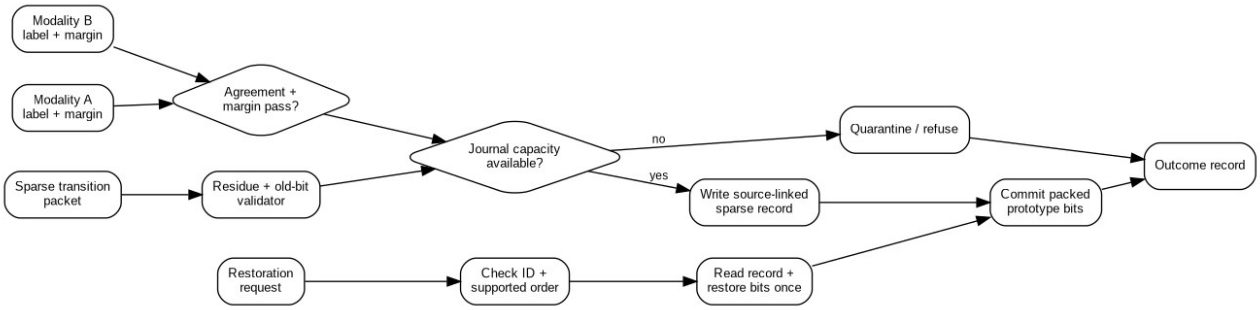
3. Monday - freeze the contract question and invariants

Monday answers the architectural question before any implementation begins: "What can one record prove and reverse?" The answer is a source-linked sparse undo record for packed prototype bits, with validation strong enough to reject malformed, stale, duplicated or misordered requests before mutation.

3.1 Invariants

- A refusal or quarantine never changes the packed prototype.
- An accepted update is journaled before prototype commit.
- Every accepted affected dimension has exactly one unique dimension_index within the packet.
- The sender's expected old/new bits are not trusted blindly; both are recomputed.
- The recomputed old bit must match the current stored prototype bit or the packet is stale.
- Restoration is exactly once and LIFO within each class.
- The journal retains enough to restore old packed bits, not hidden counts.

3.2 Architecture flow



The crucial ordering is: validate/gate -> check capacity -> write journal -> commit prototype -> emit outcome. Restoration enters through a separate ID/order check, reads the record once, restores bits, retires the record, then emits the outcome.

3.3 Pass condition

A different team should be able to take only the written contract, decode a packet, recompute old/new bits, select one deterministic outcome code, and compute wire/journal byte counts without asking the authors. The rest of the week creates the evidence for that condition.

4. Tuesday - freeze bits, bytes, signed arithmetic and sizing

4.1 Header byte offsets

Field	Byte offset	Width (bits)
source_id	0	8
update_id	1-2	16
class_id	3	8 (2 used)
sequence	4-5	16
hA	6	8
mA	7	8
hB	8	8
mB	9	8
entry_count F	10	8
length	11-12	16
reserved/format	13	8
checksum	14-15	16

R1 uses little-endian for 16-bit integers. class_id uses only bits [1:0]; bits [7:2] must be zero. reserved/format is zero in the frozen build.

4.2 Wire entry packing

```
24-bit entry E (little-endian bytes)
E[11:0]  = dimension_index
E[15:12] = pre_update_residue (signed 4-bit)
E[19:16] = update_contribution (signed 4-bit)
E[20]    = expected_old_bit
E[21]    = expected_new_bit
E[23:22] = 0
```

4.3 Journal entry packing

```
16-bit journal entry J
J[11:0]  = dimension_index
J[15:12] = pre_update_residue
```

The contribution and expected bits are discarded after they have been verified. This reduces each retained entry from 3 bytes to 2 bytes.

4.4 Signed arithmetic and the +8 representation trap

A signed 4-bit two's-complement nibble represents integers -8 through +7. Therefore the source phrase |u|<=8 cannot be interpreted as permitting wire value +8 without changing the field width/encoding. R1 allows the encoded alphabet -8..+7 and refuses a conceptual host-side +8 before packing.

```
decode4(n) = n       for 0 <= n <= 7
           = n - 16  for 8 <= n <= 15

representable set = {-8,-7,...,0,...,+6,+7}
```

Retained failure
Vector V037 uses r=7,u=+8. It is deliberately refused because +8 cannot be encoded in the 4-bit signed field. This is exactly the kind of Week 1 failure evidence that should be retained rather than silently "fixed".

4.5 Five-bit sum requirement

Although r and u are each 4-bit signed, r+u spans -16..+14. That range requires at least 5 signed bits. An RTL implementation that adds in 4 bits would wrap and can flip new_bit incorrectly.

```
min_sum = -8 + (-8) = -16
max_sum = +7 + (+7) = +14
=> signed result width >= 5 bits
```

4.6 Byte sizing and break-even proof

```
R(F) = 16 + 2F
R(40)  = 16 + 80  = 96 B
R(96)  = 16 + 192 = 208 B
R(247) = 16 + 494 = 510 B
R(248) = 16 + 496 = 512 B
```

A full 4096-bit prototype snapshot is 512 B. The sparse record is strictly smaller when:

```
16 + 2F < 512
2F < 496
F < 248
```

The source requests F=247/F=248 to expose the strict inequality. Operationally, however, the packet cap is F<=96, so the real maximum supported record is 208 B.

4.7 Wire packet length clarification

The source calls the header 16 bytes and includes the two checksum bytes in that 16-byte field table, while its note for length says header plus payload plus checksum. Counting the checksum again would contradict the 16-byte header. R1 therefore treats the listed header as checksum-inclusive:

```
L_wire(F) = 16 + 3F
L_wire(40) = 136 B
L_wire(96) = 304 B
```

This clarification is recorded in the decision log rather than being hidden as an implementation assumption.

5. Wednesday - integrate refusal logic, ordering and record lifecycle

5.1 Outcome table

Code	Meaning	Primary trigger
0	accepted	journal durable, then commit
1	quarantined	trust gate fails
2	refused-capacity	no index slot/ring span
3	refused-checksum	CRC mismatch
4	refused-residue	numeric alphabet/range invalid
5	refused-stale	recomputed old bit != stored bit
6	refused-duplicate	duplicate dimension in packet
7	refused-ordering	sequence/restore order unsupported
8	refused-bit-state	expected bits or frozen bit-format invalid
9	restored	old bits restored once
10	already-retired	repeat restore; no write

5.2 Deterministic validation precedence

1. CRC check. Mismatch -> 3.
2. Frozen structural checks (class/reserved/F/length). Malformed bit-format -> 8.
3. Numeric alphabet/range check -> 4.
4. Duplicate dimension check -> 6.
5. Recompute expected old/new bits -> mismatch -> 8.
6. Trust gate hA==hB==class_id and both margins -> otherwise quarantine 1.
7. Per-class sequence check -> 7.
8. Current packed old-bit check -> stale 5.
9. Journal slot/ring capacity -> 2.
10. Write journal record, then commit prototype -> 0.

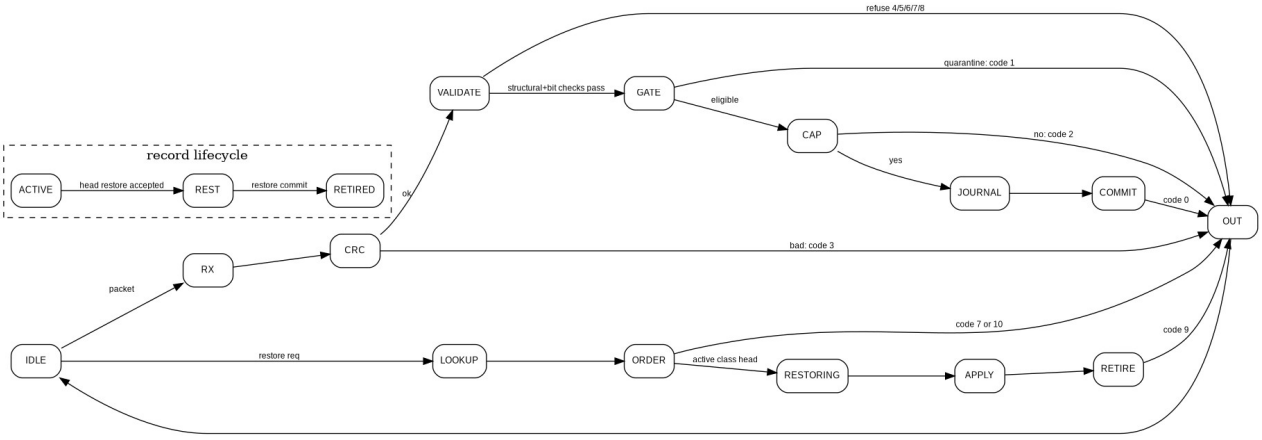
The precedence matters because one malformed packet may violate more than one rule. A frozen priority guarantees one reproducible primary outcome.

5.3 Agreement and margin gate

```
eligible = (hA == hB == class_id)
           and (mA >= theta_A)
           and (mB >= theta_B)
```

theta_A and theta_B are empirical. The source explicitly says the example value 128 is illustrative, so this package leaves the actual thresholds null until calibration evidence is supplied. A failure here is quarantine, not a protocol refusal.

5.4 Ordering state diagram and lifecycle



Each class owns a head pointer to its newest ACTIVE record. Only the head may restore, giving reverse commit order within the class. A successful request moves ACTIVE -> RESTORING -> RETIRED. A second request against a retained retired tombstone returns code 10 and performs no write.

5.5 Ring wrap choice

R1 forbids splitting one journal record across the physical ring boundary. If fewer than R(F) bytes remain to the end, the allocator skips/pads the tail and tries offset 0. This costs at most Rmax-1 = 207 bytes of tail space at a wrap but makes every record a single contiguous base+length span and simplifies hardware.

5.6 Stream order and 64-bit word grouping

The record preserves sender stream order, but prototype memory updates are combined by 64-bit word. For dimension d:

```
word_index = d >> 6
bit_index  = d & 63
```

All touched bits of a word are accumulated into a mask/value pair so one grouped prototype update is performed for that word. Because duplicate dimensions are refused, no bit position can receive conflicting assignments within a packet.

6. Thursday - evidence: worked vectors, capacity and consistency checks

6.1 Worked vector method

Each valid vector supplies r and u, computes the exact 5-bit-capable integer sum, then applies sign_threshold to r and r+u. Thirty-six valid vectors are included, exceeding the minimum of thirty. Six extra refusal examples preserve boundary/failure cases.

ID	r	u	r+u	old	new	verdict
V001	-8	-8	-16	0	0	accept
V002	-8	-1	-9	0	0	accept
V003	-8	0	-8	0	0	accept
V004	-8	7	-1	0	0	accept
V005	-7	-1	-8	0	0	accept
V006	-7	7	0	0	1	accept
V007	-6	6	0	0	1	accept
V008	-5	-3	-8	0	0	accept
V009	-5	5	0	0	1	accept
V010	-5	7	2	0	1	accept
V011	-4	-4	-8	0	0	accept
V012	-4	3	-1	0	0	accept
V013	-4	4	0	0	1	accept
V014	-3	-5	-8	0	0	accept
V015	-3	2	-1	0	0	accept
V016	-3	3	0	0	1	accept
V017	-2	-6	-8	0	0	accept
V018	-2	1	-1	0	0	accept

The complete 42-vector table appears in Appendix 14 and in tests/hand_vectors.csv.

6.2 Boundary examples

(r,u)	sum	bits/verdict	Why useful
(-8,-8)	sum=-16	0 -> 0	lowest possible valid sum
(-7,+7)	sum=0	0 -> 1	crosses exactly onto zero tie
(0,-1)	sum=-1	1 -> 0	zero old state crosses negative
(7,-8)	sum=-1	1 -> 0	largest residue with most-negative contribution
(7,+7)	sum=14	1 -> 1	largest valid positive sum
(7,+8)	N/A	refuse	+8 not encodable in signed 4-bit

6.3 Record capacity math

R1 selects a 256 KiB journal ring and 1024 index slots. At the worst supported record size 208 B:

```
1024 * 208 B = 212,992 B
256 KiB = 262,144 B
headroom = 49,152 B
ideal max-size physical records = floor(262,144/208) = 1,260
```

Therefore the index depth N=1024, not raw ring bytes, intentionally limits simultaneously indexable live records. slot_id width is ceil(log2(1024))=10 bits.

6.4 KV260 memory sanity check

AMD DS986 lists 144 Block RAM blocks on the KV260. Treating one BRAM36 as 36 Kib, the selected journal uses approximately:

```
262,144 B * 8 / (36 * 1024) = 56.9 BRAM36 equivalents
```

A reference 4096-entry trace at 16 B/entry is 64 KiB (about 14.2 BRAM36 equivalents). A reference 1024-entry index at 16 B/entry is 16 KiB (about 3.6 BRAM36 equivalents). The combined payload is about 336 KiB, or 74.7 BRAM36 equivalents before synthesis overhead. This is a feasibility sanity check, not a post-synthesis utilization claim.

6.5 Performance target sanity check

At 100 MHz, a 50,000 packet-decision/s target yields 2,000 clock cycles per packet on average. The Week 1 brief does not require a timed run, so this is deliberately a design target to be measured later, not a claimed throughput result.

```
cycles_per_packet_budget = 100,000,000 / 50,000 = 2,000 cycles
```

6.6 What still requires human/empirical evidence

- Name the exact encoder/dataset and split.
- Measure actual per-dimension contributions and show the frozen alphabet is not exceeded.
- Calibrate theta_A and theta_B from validation data.
- Have a second person independently check the vector file and sign/date the review.

7. Friday - evaluation, numeric result and Green/Amber conditions

Numeric result
Maximum supported journal record size = 208 bytes.

7.1 Friday review answers

Review item	Recorded answer	Status
Maximum supported record size	208 B	complete
Break-even proof	F<248	complete
Outcome table	0..10 with triggers	complete
16-item decision log	all recorded	complete; empirical items owned/open
Worked vectors	42 total, 36 valid	complete
Source/build ID	source SHA-256 captured; Git commit missing	external
Encoder/dataset	not in uploaded source	external
theta_A/theta_B	not calibrated here	external
Independent reviewer	not possible to fabricate	external

7.2 Why the package is AMBER rather than falsely GREEN

The technical contract is complete, but a Green checkpoint in the supplied brief also depends on real team evidence. The encoder/dataset, empirical contribution distribution, calibrated thresholds, Git commit and independent reviewer are not present in the uploaded file. Those cannot be honestly manufactured. The correct status is therefore AMBER pending external evidence, with explicit conditions to turn it Green.

7.3 Green closure checklist

- Fill encoder + dataset identity/revision/split.
- Attach real contribution-distribution evidence and confirm no alphabet violations, or redesign the field/alphabet together.
- Fill calibrated theta_A/theta_B and calibration method.
- Replace source_commit placeholder with the actual repository commit.
- Obtain independent vector/document countersignature.
- Cross-check state labels against the actual PROJECT_PLAN Section 3.3.

8. Exact wire packet and journal record specification

8.1 Header

Field	Offset	Bytes	Width	Rule
source_id	0	1	8	source/harness ID
update_id	1-2	2	16	global update ID
class_id	3	1	8	low 2 bits; high bits 0
sequence	4-5	2	16	per-class next sequence
hA,mA	6-7	2	8+8	A label/margin
hB,mB	8-9	2	8+8	B label/margin
entry_count F	10	1	8	0..96
length	11-12	2	16	16+3F wire / 16+2F journal
reserved/format	13	1	8	0
checksum	14-15	2	16	CRC-16 field

8.2 Wire entry exact bits

```
23              22 21 20 19          16 15          12 11                  0
+----------------+--+--+---------------+--------------+----------------------+
| reserved = 0   |NB|OB| contribution  | pre-residue  | dimension_index      |
+----------------+--+--+---------------+--------------+----------------------+
  2 bits           1  1      4 signed        4 signed          12 bits
```

8.3 Journal entry exact bits

```
15          12 11                  0
+--------------+----------------------+
| pre-residue  | dimension_index      |
+--------------+----------------------+
   4 signed             12 bits
```

8.4 Header reuse clarification for persisted records

R1 interprets "reuses the 16-byte header unchanged" as reusing the schema. For a persisted journal record, length is set to 16+2F and checksum is recomputed over the journal representation. All identity, class, sequence and modality fields are copied. This makes a journal record self-contained and checksum-verifiable.

9. Validation equations and deterministic refusal precedence

9.1 Three distinct bit comparisons

There are three conceptually different checks and they must not be collapsed:

11. Sender consistency: expected_old_bit must equal sign_threshold(r).
12. Sender transition consistency: expected_new_bit must equal sign_threshold(r+u).
13. Freshness: recomputed old_bit must equal the currently stored prototype bit.

The first two detect malformed or dishonest packet bit claims (code 8). The third detects a packet created from an outdated prototype state (code 5).

9.2 Example: stale versus bit-state

Suppose r=-1, u=+1. R1 computes old=0, sum=0, new=1. If the packet claims expected_old=1, it is code 8 immediately. If the packet claims expected_old=0/new=1 correctly but the current prototype bit is now 1 because another accepted update already changed it, the packet is code 5 stale.

9.3 Structural format checks

- class_id high six bits zero and numeric class in 0..3;
- F in 0..96;
- length equals the exact representation formula;
- reserved/format byte zero;
- entry reserved bits zero.

Because the frozen outcome list has no dedicated format/length code, R1 maps these frozen bit-format failures to code 8 and records the choice in the decision log.

9.4 Atomic commit rule

```
VALIDATE ALL -> ALLOCATE -> WRITE JOURNAL -> VERIFY/FINALIZE JOURNAL -> COMMIT PROTOTYPE ->
OUTCOME 0
```

This ordering ensures that any prototype mutation always has a prior durable undo record. A crash/fault model is outside Week 1, but this ordering is the minimum contract needed for later transactional implementation.

10. Restoration algorithm and 64-bit write-mask mathematics

10.1 Record lookup and ordering

14. Find update_id in the record index table.
15. If state is RETIRED, return code 10.
16. Verify slot is the current class head; otherwise return code 7.
17. Transition ACTIVE -> RESTORING to block a second concurrent restore.
18. Read the contiguous journal record and verify its checksum.
19. For every 2-byte entry, decode dimension d and residue r, compute old_bit=sign_threshold(r).
20. Accumulate per-word mask/value pairs and apply one masked update per touched 64-bit word.
21. Transition RESTORING -> RETIRED, move the class head to the prior record, emit code 9.

10.2 Mask equations

```
w = d >> 6
b = d & 63
mask[w]  = mask[w] | (1 << b)
value[w] = (value[w] & ~(1 << b)) | (old_bit << b)
```

At commit/restore time, the prototype word update is conceptually:

```
new_word = (old_word & ~mask[w]) | (value[w] & mask[w])
```

This formulation touches only affected dimensions. The duplicate-dimension refusal rule guarantees each bit has a single intended value.

10.3 Why reverse order is required

If two accepted updates touch the same bit, the older record's residue describes the bit state before the older update, not the state after the newer update. Restoring the older record first can erase the effect needed by the newer record's history. Enforcing per-class LIFO ensures each record is applied to the state it was designed to reverse.

11. Journal, index and trace sizing

11.1 Index depth and slot width

```
N = 1024
slot_id_bits = ceil(log2(N)) = ceil(log2(1024)) = 10
```

N is the maximum simultaneously indexable live records across all four classes. A retired tombstone may remain until its slot is safely reused; repeat restore detection is guaranteed only while that index identity remains retained.

11.2 Reference memory budget

Memory	Size	Purpose
Journal ring	256 KiB	sparse record bytes
Trace	4096 x 16 B = 64 KiB	one reference demonstration window
Index	1024 x 16 B = 16 KiB	reference packed metadata budget
Total payload	336 KiB	before control/mask overhead

11.3 Trace reference entry

The brief fixes trace depth but not trace field format. The R1 sizing assumption uses a 16-byte trace entry so depth can be translated to bytes. A practical 16-byte entry can hold a cycle/timestamp, update_id, sequence, class, outcome, F/source flags and a 32-bit journal offset. This trace shape is a reference sizing aid, not a source-frozen interface.

11.4 Worst-case ring wrap waste

With no-split records and Rmax=208 B, a wrap can discard at most 207 trailing bytes. The selected 49,152 B ring headroom over 1024 maximum-size records is far larger than one wrap pad event; actual allocator safety must still account for the live tail/head geometry rather than rely only on aggregate free bytes.

12. CRC and AXI-Lite integration profile

12.1 CRC-16/CCITT-FALSE

Parameter	R1 value
Polynomial	0x1021
Initial value	0xFFFF
Reflect input	false
Reflect output	false
XOR out	0x0000
Checksum field during calculation	zero

The same algorithm is used for wire and persisted representations. Compute with checksum bytes zero, then store the result. Validation repeats zero-and-recompute and compares against the field.

12.2 AXI-Lite project allocation

```
Base address : 0xA0000000
Span         : 0x00010000 (64 KiB)
Path         : PS M_AXI_HPM0_FPD -> SmartConnect -> REVOKE-HDC AXI4-Lite slave
```

AMD PG201 identifies M_AXI_HPM0_FPD/HPM1_FPD as processing-system master interfaces to programmable-logic slaves. The exact address is a project allocation, not a board-wide immutable address; Vivado Address Editor must be treated as authoritative and must match the software device-tree/driver view.

12.3 100 MHz / 50k decisions per second

These are R1 design targets to close the Week 1 decision, not benchmark claims. Later RTL weeks should report measured initiation interval/latency and replace the target-only statement with empirical throughput.

13. All sixteen Week 1 decisions

#	Topic	Status	Answer	Owner
1	Encoder/dataset	OPEN evidence	Name exact encoder + dataset/split; not present in upload.	HDC/ML
2	Residue/alphabet	Contract closed	r,u encoded as signed 4-bit -8..+7; empirical distribution still required.	HDC/ML
3	Duplicates	Confirmed	Refuse entire packet, code 6.	System
4	Entry ordering	Decided	Preserve stream order; group writes by word.	System
5	Write-mask	Confirmed	One mask/value group per touched 64-bit word.	RTL
6	theta A/B	OPEN empirical	Calibrate; do not claim illustrative 128.	HDC/ML
7	Journal allocation	Confirmed	Ring + index + per-class heads.	System
8	Ring wrap	Decided	No split; pad to end then restart at 0.	RTL
9	Restore order	Confirmed	Only active class head.	System
10	Break-even	Closed	F<248; F workload distribution later.	Reproducer
11	Clock/rate	Target set	100 MHz; >=50k packet decisions/s target.	FPGA
12	Journal BRAM	Decided	256 KiB ring.	FPGA
13	Index depth N	Decided	1024 slots, slot_id 10 bits.	System
14	Trace depth	Decided	4096 entries; reference 16 B/entry.	Reproducer
15	CRC	Decided	CRC-16/CCITT-FALSE.	System
16	AXI-Lite	Decided	0xA0000000 / 64 KiB via HPM0 FPD.	FPGA integration

The full decision log in the package contains longer rationale, ownership and source-vs-R1 distinctions.

14. Worked vector appendix

Rows V001-V036 are valid alphabet vectors. V037-V042 are explicit refusals. Empty sum/bit cells mean the value is rejected before a valid encoded arithmetic operation exists.

ID	r	u	sum	old	new	verdict	reason
V001	-8	-8	-16	0	0	accept	r and u are representable; threshold arithmetic shown
V002	-8	-1	-9	0	0	accept	r and u are representable; threshold arithmetic shown
V003	-8	0	-8	0	0	accept	r and u are representable; threshold arithmetic shown
V004	-8	7	-1	0	0	accept	r and u are representable; threshold arithmetic shown
V005	-7	-1	-8	0	0	accept	r and u are representable; threshold arithmetic shown
V006	-7	7	0	0	1	accept	r and u are representable; threshold arithmetic shown
V007	-6	6	0	0	1	accept	r and u are representable; threshold arithmetic shown
V008	-5	-3	-8	0	0	accept	r and u are representable; threshold arithmetic shown
V009	-5	5	0	0	1	accept	r and u are representable; threshold arithmetic shown
V010	-5	7	2	0	1	accept	r and u are representable; threshold arithmetic shown
V011	-4	-4	-8	0	0	accept	r and u are representable; threshold arithmetic shown
V012	-4	3	-1	0	0	accept	r and u are representable; threshold arithmetic shown
V013	-4	4	0	0	1	accept	r and u are representable; threshold arithmetic shown
V014	-3	-5	-8	0	0	accept	r and u are representable; threshold arithmetic shown
V015	-3	2	-1	0	0	accept	r and u are representable; threshold arithmetic shown
V016	-3	3	0	0	1	accept	r and u are representable; threshold arithmetic shown
V017	-2	-6	-8	0	0	accept	r and u are representable; threshold arithmetic shown

ID	r	u	sum	old	new	verdict	reason
V018	-2	1	-1	0	0	accept	r and u are representable; threshold arithmetic shown
V019	-2	2	0	0	1	accept	r and u are representable; threshold arithmetic shown
V020	-1	-7	-8	0	0	accept	r and u are representable; threshold arithmetic shown
V021	-1	0	-1	0	0	accept	r and u are representable; threshold arithmetic shown
V022	-1	1	0	0	1	accept	r and u are representable; threshold arithmetic shown
V023	0	-8	-8	1	0	accept	r and u are representable; threshold arithmetic shown
V024	0	-1	-1	1	0	accept	r and u are representable; threshold arithmetic shown
V025	0	0	0	1	1	accept	r and u are representable; threshold arithmetic shown
V026	0	7	7	1	1	accept	r and u are representable; threshold arithmetic shown
V027	1	-8	-7	1	0	accept	r and u are representable; threshold arithmetic shown
V028	1	-1	0	1	1	accept	r and u are representable; threshold arithmetic shown
V029	1	0	1	1	1	accept	r and u are representable; threshold arithmetic shown
V030	2	-3	-1	1	0	accept	r and u are representable; threshold arithmetic shown
V031	2	-2	0	1	1	accept	r and u are representable; threshold arithmetic shown
V032	2	5	7	1	1	accept	r and u are representable; threshold arithmetic shown
V033	4	-5	-1	1	0	accept	r and u are representable; threshold arithmetic shown
V034	4	-4	0	1	1	accept	r and u are representable; threshold arithmetic shown
V035	7	-8	-1	1	0	accept	r and u are representable; threshold arithmetic shown

ID	r	u	sum	old	new	verdict	reason
V036	7	7	14	1	1	accept	r and u are representable; threshold arithmetic shown
V037	7	8		1		refuse	conceptual +8 passes |u|<=8 wording but is not encodabl...
V038	0	9		1		refuse	contribution exceeds alphabet and is not encodable
V039	0	-9		1		refuse	contribution magnitude/alphabet exceeded
V040	8	0				refuse	pre_update_residue outside signed 4-bit range
V041	-9	0				refuse	pre_update_residue outside signed 4-bit range
V042	3	12		1		refuse	contribution exceeds alphabet and is not encodable

15. Ambiguities discovered and how R1 resolves them

Issue	Why it matters	R1 resolution
Checksum counted twice?	The 16-byte header table already includes checksum, while the length note says header + payload + checksum.	R1: checksum is inside the 16 bytes; L_wire=16+3F.
Can +8 fit in signed 4-bit?	No; two's-complement range is -8..+7.	R1: encoded contribution set is -8..+7; conceptual +8 is refused before packing.
What is sign_threshold(0)?	Not defined in the uploaded file.	R1: freeze zero as 1 (x>=0).
Does journal "reuse header unchanged" include length/checksum bytes?	Literal reuse would leave wire length/CRC inconsistent with the 2-byte journal entries.	R1: reuse schema; recompute journal length and checksum.
No dedicated format refusal code	Outcome list has checksum/residue/stale/duplicate/ordering/bit-state but no format/length.	R1: frozen bit-format violations map to code 8.
Exact controller state names	PROJECT_PLAN Section 3.3 not supplied.	R1: use states derived from the supplied flow; rename after cross-check without changing semantics.

Recording these clarifications is safer than silently choosing an interpretation. It also makes future changes auditable: if the team later decides a different zero tie or journal checksum behavior, it can version the contract and regenerate vectors rather than drifting implementation-by-implementation.

16. Week 2 hand-off

Week 2 can now build a software reference model against a deterministic data model. The smallest next test specified by the brief is one hand-built two-entry packet that is admitted, journaled, committed and then restored.

16.1 Recommended two-entry test

Choose two dimensions in different 64-bit words so the software model exercises two masks. Example contract-level values:

Entry	dimension	residue	contribution	old	new
Entry 0	d=5	r=-1	u=+1	old=0	new=1
Entry 1	d=130	r=+2	u=-3	old=1	new=0

Before commit, the current prototype must contain bit5=0 and bit130=1. After applying the update it should contain bit5=1 and bit130=0. Restoration from the journal residues must return bit5=0 and bit130=1 exactly once.

16.2 Week 2 assertions inherited from Week 1

- Wire parser exactly matches the frozen byte/bit layout.
- All arithmetic uses signed decode and >=5-bit sums.
- Outcome precedence matches the Week 1 table.
- Journal record is written before prototype mutation.
- Restoration enforces active class head and already-retired behavior.
- Software model consumes actual calibrated thresholds once the team supplies them.

17. References and submission checklist

17.1 Supplied project source

WEEK_1.md - P3 REVOKE-HDC: Week 1 - Freeze update alphabet, residue and record format. Supplied as the task basis. SHA-256: 38992d4c044d56fda32810f798eb3dccc67e03fb975457de48fc86507cfbb4aa.

17.2 External platform references

22. AMD Kria KV260 Vision AI Starter Kit Data Sheet DS986, Product Details. https://docs.amd.com/r/en-US/ds986-kv260-starter-kit/Product-Details
23. AMD Kria KV260 Vision AI Starter Kit User Guide UG1089, Summary. https://docs.amd.com/r/en-US/ug1089-kv260-starter-kit/Summary
24. AMD Zynq UltraScale+ MPSoC Processing System Product Guide PG201, PS-PL AXI interfaces. https://docs.amd.com/r/en-US/pg201-zynq-ultrascale-plus-processing-system/Slave-Interface

17.3 Deliverable checklist

Item	Status
Bit-field tables in docs	DONE
Versioned config mirror	DONE
Validation equations + stale rule	DONE
All 16 decision items recorded	DONE
Ordering/lifecycle diagram	DONE
Sizing sheet with F=40/96/247/248	DONE
>=30 valid vectors	DONE - 36 valid
Outcome/refusal table	DONE
Manifest	DONE
Friday numeric result	DONE - 208 B
Actual encoder/dataset evidence	TEAM INPUT
Calibrated thresholds	TEAM INPUT
Git commit	TEAM INPUT
Independent reviewer signature	TEAM INPUT

Final submission note
The package is suitable as a complete Week 1 reference solution and submission scaffold. To claim the source brief's fully measured Green checkpoint, fill the four real-world evidence items above rather than replacing them with invented values.